



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

EndSemester

Semester: III

Programme: B. Tech Information Technology

Course: Data Structure

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT32301	Course Outcome
CO1	To gain knowledge about basic concepts of data structures.
CO2	To acquire knowledge of stacks and queues with their applications.
CO3	To aware about the concepts of trees with their applications.
CO4	To learn and design the algorithm for graphs with their applications.
CO5	To implement basic operations on linked list.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	What is Selection Sort?	CO1	2	2
b.	Define Empty List.	CO2	2	2
c.	What is Priority Queue?	CO3	2	2
d.	Define Graph Traversal.	CO4	3	2
e.	What is AVL Tree?	CO5	3	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain Merge Sort with suitable example.	CO1	3	5
b.	Differentiate between Linear Search and Binary Search.	CO1	3	5
c.	Explain Quick Sort algorithm and its advantages.	CO1	4	5
Que. 3	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain insertion operation in Linked List with example.	CO2	4	5
b.	Explain deletion operation in Circular Linked List.	CO2	3	5
c.	Compare Array and Linked List.	CO2	2	5
Que. 4	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Stack operations Push and Pop with example.	CO3	2	5
b.	Explain Circular Queue implementation.	CO3	4	5
c.	Describe applications of Stack in expression conversion.	CO3	3	5
Que 5	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain graph representation using Adjacency Matrix.	CO4	2	5
b.	Explain BFS traversal with suitable example.	CO4	3	5
c.	Explain Prim's Algorithm with example.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Binary Search Tree with operations.	CO5	2	5
b.	Explain Binary Tree Traversals.	CO5	3	5
c.	Explain AVL Tree balancing operations.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create